

System: GridForge Plant Operations Platform

Classification: Confidential

Deployment: Hybrid

Regulatory Scope: NIST, ISO 27001

Trust Boundary: External Access Zone

Contains:

- Supplier Portal User [external_entity]
- Remote Maintenance Vendor [external_entity]
- Site Reliability Analyst [external_entity]
- VPN Gateway [process]

Trust Boundary: Plant DMZ

Contains:

- OT Jump Host [process]
- Maintenance Orchestrator [process]
- Telemetry Broker [process]
- Predictive Analytics Service [process]

Trust Boundary: OT Core Zone

Contains:

- Field Gateway [process]
- Plant Historian [data_store]
- PLC Command Store [data_store]

Data Flows:

- Remote Maintenance Vendor -> VPN Gateway: remote maintenance tunnel
- Site Reliability Analyst -> VPN Gateway: emergency operator session
- VPN Gateway -> OT Jump Host: interactive privileged session
- OT Jump Host -> Maintenance Orchestrator: approved maintenance task
- Maintenance Orchestrator -> Field Gateway: maintenance job dispatch
- Field Gateway -> Telemetry Broker: plant telemetry stream
- Telemetry Broker -> Plant Historian: time-series archive write
- Predictive Analytics Service -> Plant Historian: analytics query
- Maintenance Orchestrator -> PLC Command Store: command package staging
- Supplier Portal User -> Maintenance Orchestrator: supplier work order upload